

System and Surrounding

→ The portion of physical universe that is under thermodynamical consideration is called system. Anything other than system is surrounding.

→ (i) Open System $\Rightarrow$ System and Surrounding can exchange both materials and energy

(ii) Closed System $\Rightarrow$ " " " " only exchange energy

(iii) Isolated System $\Rightarrow$ " " " " exchange nothing.

Standard Enthalpy Change of Formation, ΔH_f°

Standard Condition

(i) Temperature :- $25^\circ\text{C} / 298\text{K}$

(ii) Pressure :- $1\text{ atm} / 10^5\text{ Pa}$

(iii) Normal state of elements and compounds.

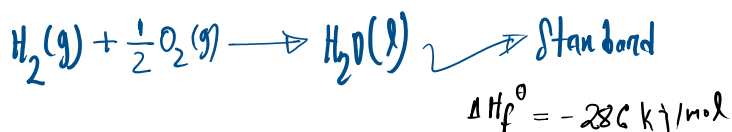
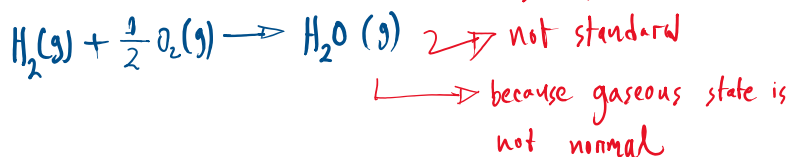
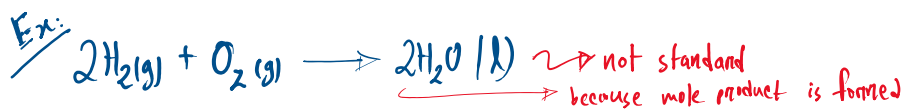
→ Energy

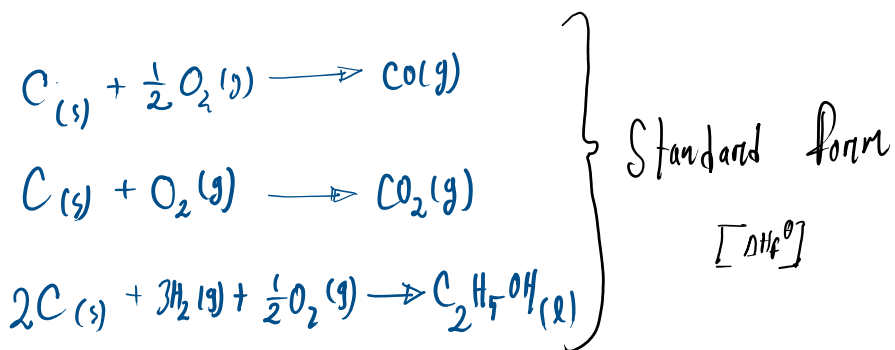
$$\Delta H = \Delta E + p\Delta V$$

→ Enthalpy

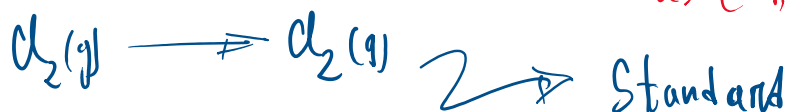
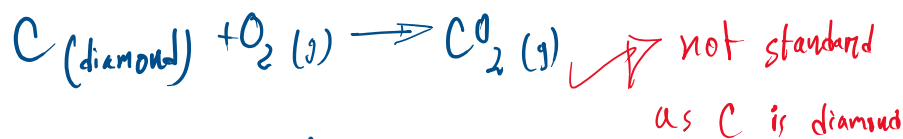
→ Pressure
Volume
Work

→ is the enthalpy change when 1 mol of a substance is from its elements under consideration.





But



$[\Delta H_f^\ominus]$ of elements is always zero, because there is no change.

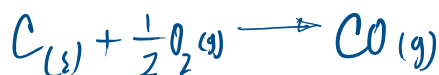
Standard Enthalpy change of combustion, $\Delta H_c^\ominus$

$\hookrightarrow$ is the enthalpy change when 1.0 mole of a substance is burnt completely in excess oxygen under standard conditions.

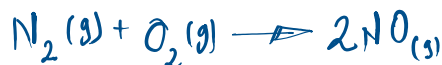
$\hookrightarrow \Delta H_c^\ominus$ is always exothermic.



$\hookrightarrow \Delta H_c^\ominus(\text{C}) = \Delta H_f^\ominus(\text{CO}_2) \leadsto \text{negative}$

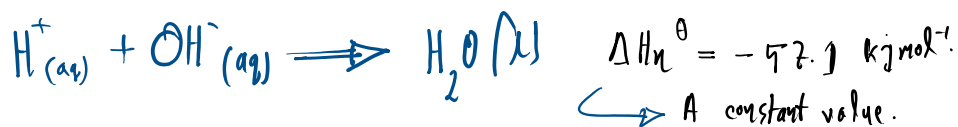


$\hookrightarrow$ not in $\Delta H_c^\ominus$, but in $\Delta H_f^\ominus$

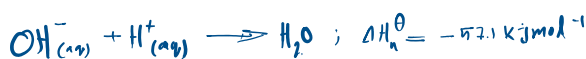
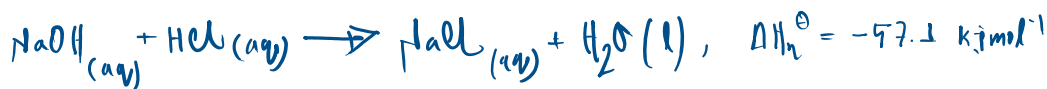


$\hookrightarrow \Delta H = \text{positive}$, so not in $\Delta H_c^\ominus$.

Standard Enthalpy Change of Neutralisation, ΔH_n^θ .



Standard enthalpy change of neutralisation is the enthalpy change when one mole water is formed from aqueous solution of acid and alkali under standard conditions



Determination of ΔH_n^θ

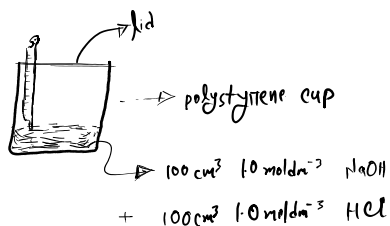
Initial Temperature = $T_1^\circ\text{C}$

Final Temperature = $T_2^\circ\text{C}$

$$\Delta T = (T_2 - T_1)^\circ\text{C}$$

$$\text{Heat change, } \Delta Q = mc\Delta T$$

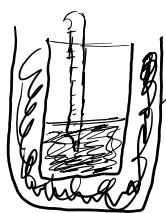
$$\begin{aligned} \text{no. of mol} &= \frac{100 \times 1.0}{1000} \\ &= 0.1 \text{ mol} \end{aligned}$$



$$\left| \begin{array}{l} m = \text{mass of solution} \\ c = \text{specific heat} \\ = 4.18 \text{ J kg}^{-1} \text{ K}^{-1} \end{array} \right.$$

0.1 mol of water formed and produces Q J heat

$$\frac{Q_{x1}}{0.1} = \Delta H_c^\circ$$



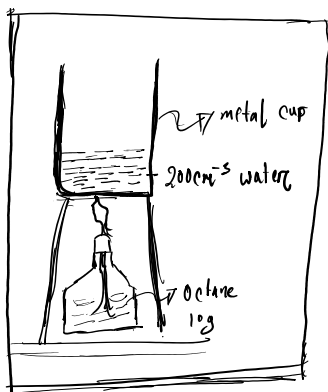
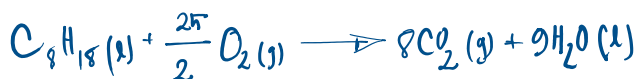
putting in a bigger cup
and surrounded by cotton

↳ for better calculation,

$$\Delta Q + Q_c = mc\Delta T$$

Heat absorbed by calorimeter setup

$C_8H_{18} \rightarrow$ Octane



heat (q) absorbed by water

$\approx$ heat produced by burning of octane

$$Q = Q_1 + Q_2 \quad \left| \begin{array}{l} Q_1 = mc\Delta T \\ Q_2 = m_2c\Delta T \end{array} \right.$$

$$= 200 \times 4.18 \times \Delta T + Q_2$$

$Q_2 \approx$ heat absorbed by calorimeter

$$C_8H_{18} \Rightarrow 8 \times 12 + 18 \Rightarrow 114 \text{ gm} \Rightarrow 1 \text{ mol}$$

$$10 \text{ gm} \Rightarrow \frac{1}{11.4} \text{ mol}$$

$$\text{for } \frac{1}{11.4} \text{ mol}, Q = 836\Delta T + m_2c$$

$$\text{for } 1 \text{ mol}, Q = 114(836\Delta T + m_2c) \text{ Joules}$$

⊗ Why the experimental value will always be lower than the real value?

Answer:

Answer:

⇒ Heat loss to surroundings

⇒ Burning of fuel will be incomplete

⇒ fuel might contain some impurity.

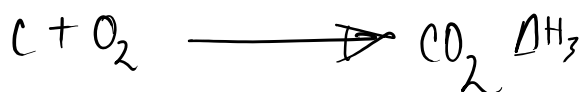
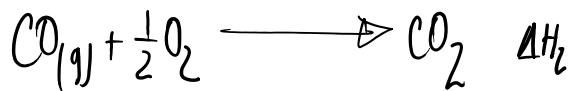
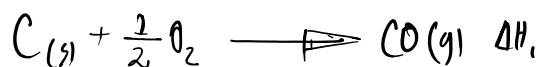
Laws of Thermochemistry

(i) Lavoisier and Laplace Law

→ The enthalpy change accompanying in one direction is exactly equal in magnitude but opposite in sign to that associated with the same reaction in reverse reaction.

(ii) Hess's Law

→ The total enthalpy change of a chemical reaction is independent of the road provided by the initial function and final condition are the same.

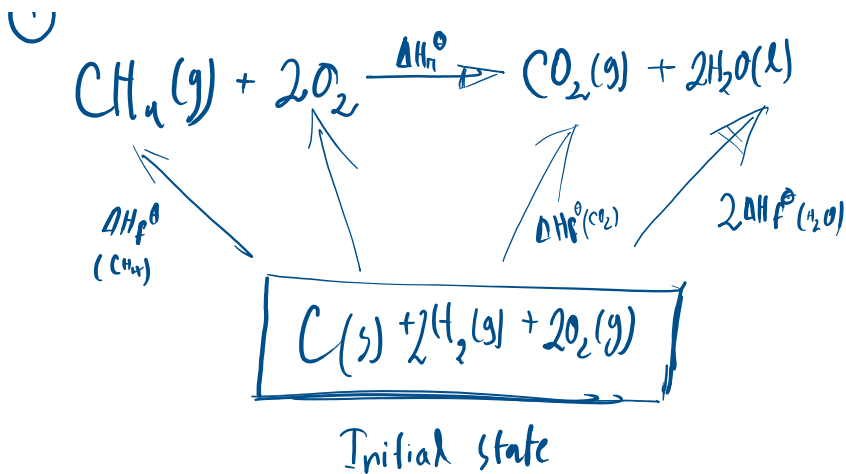


$$\Delta H_3 = \Delta H_1 + \Delta H_2$$

Hess's cycle

(1)



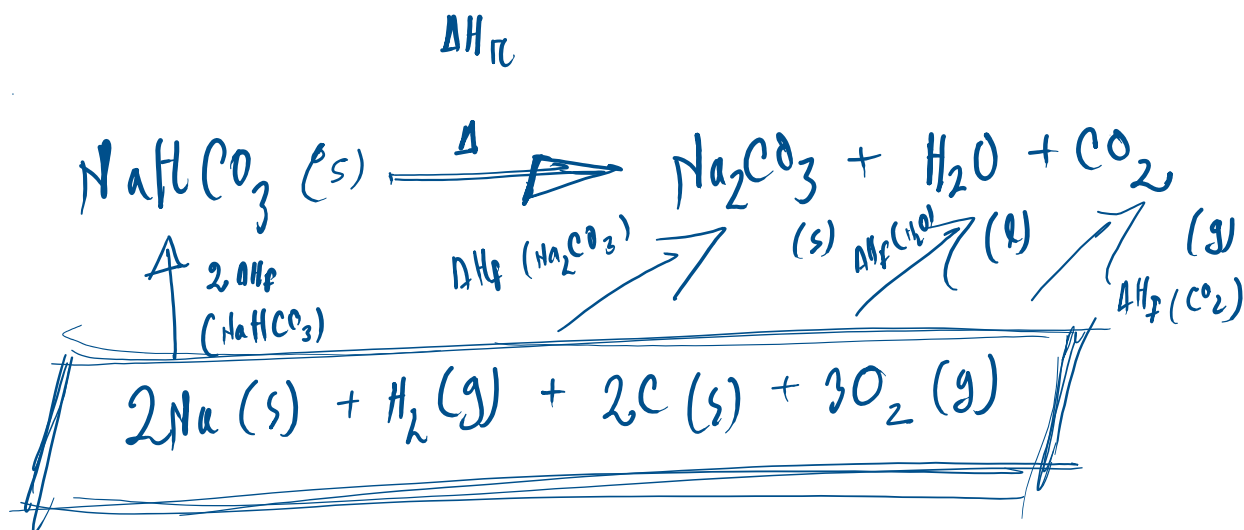


$$\Delta H_f(\text{CH}_4) + \Delta H_r^\circ = \Delta H_f^\circ(\text{CO}_2) + 2\Delta H_f^\circ(\text{H}_2\text{O})$$

$$\Delta H_r = \Delta H_f(\text{CO}_2) + 2\Delta H_f(\text{H}_2\text{O}) - \{ \Delta H_f(\text{CH}_4) \}$$

ΔH_r = formation enthalpy of products $\uparrow$ reactants
 - formation enthalpy of reactants

②

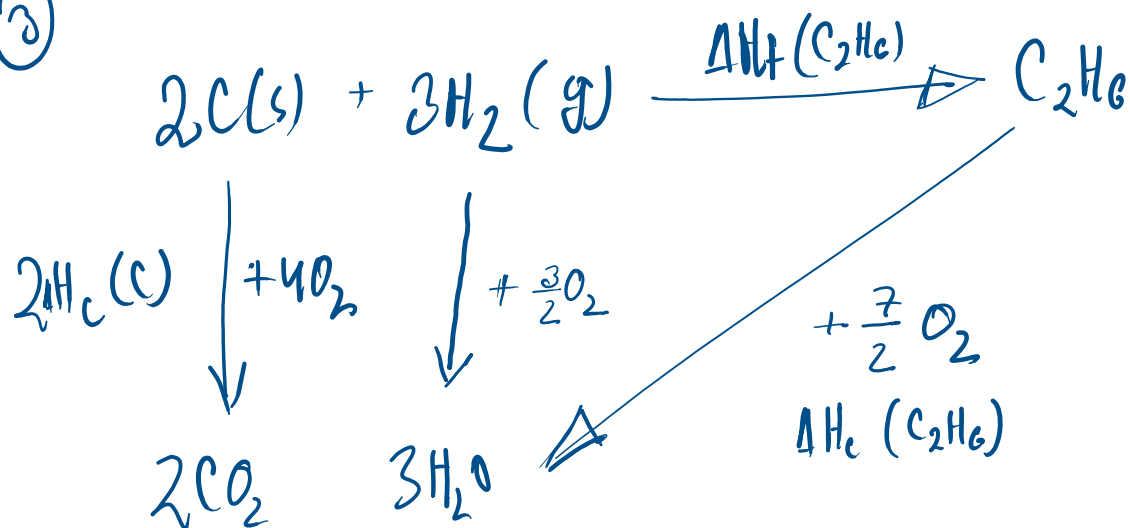


$$2\Delta H_f(\text{NaHCO}_3) + \Delta H_r = \Delta H_f(\text{Na}_2\text{CO}_3) + \Delta H_f(\text{H}_2\text{O}) + \Delta H_f(\text{CO}_2)$$

ΔH_r = formation enthalpy of products - formation enthalpy of reactants

ΔH_c = formation enthalpy of products - formation enthalpy of reactants.

③



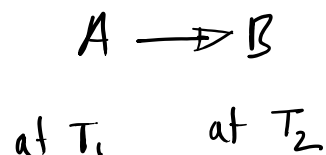
$$\Delta H_f(C_2H_6) + \Delta H_c(C_2H_6) = 2\Delta H_c(C) + 3\Delta H_c(H_2)$$

$$\Delta H_f(C_2H_6) = 2\Delta H_c(C) + 3\Delta H_c(H_2) - \Delta H_c(C_2H_6)$$

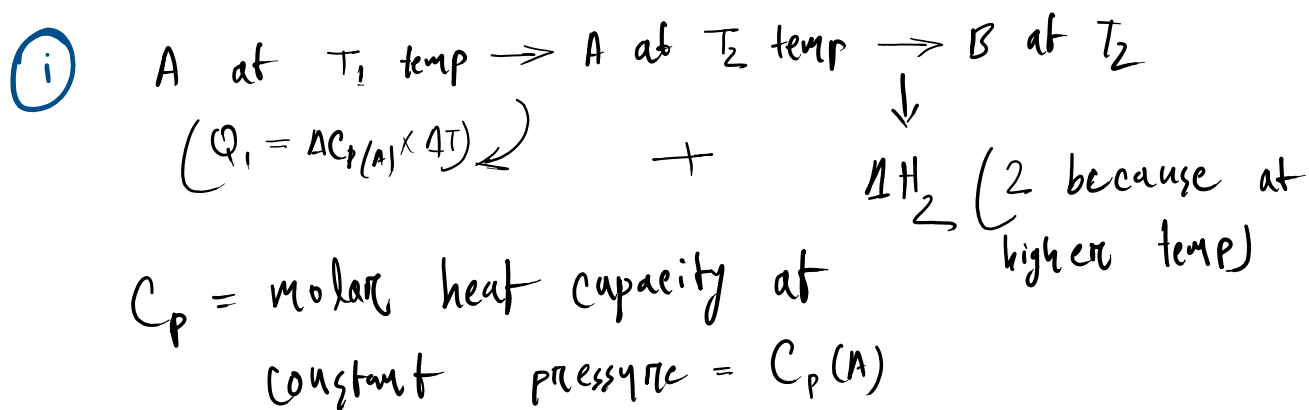
ΔH_c = combustion enthalpy of reactants -
combustion enthalpy of products

Effect of temperature of ΔH (Kirchoff's Equation)

Let us consider a reaction

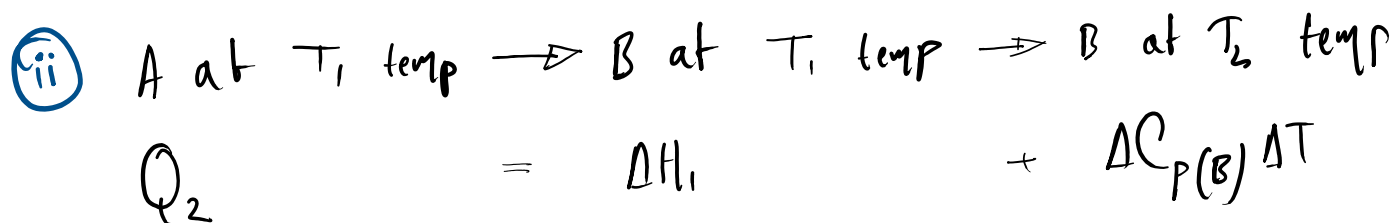


This process can be carried out in two ways:



$$\Delta T = T_2 - T_1$$

Total enthalpy change = Q_1



Total enthalpy change = $Q_2 = \Delta H_1 + \Delta C_{p(B)} \Delta T$

According to Hess's law,

$$Q_1 = Q_2$$

$$\Rightarrow \Delta H_2 + \Delta C_p(A) \Delta T = \Delta H_1 + \Delta C_p(B) \Delta T$$

$$\Rightarrow \Delta H_2 - \Delta H_1 = (\Delta C_p(B) - \Delta C_p(A)) \Delta T$$

$$\Rightarrow \frac{\Delta H_2 - \Delta H_1}{\Delta T} = \Delta C_p(B) - \Delta C_p(A)$$

$$\Rightarrow \frac{d\Delta H}{dT} = \Delta C_p$$

$\Rightarrow$ Kirchhoff's equation

	Problem solved	
1.	Calculate the enthalpy change of the reaction, ΔH_r°, $\text{C}_2\text{H}_4(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{C}_2\text{H}_6(\text{g})$ Given data are as follows $\text{C}_2\text{H}_4(\text{g}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) \quad \Delta H_r^\circ = -1401 \text{ kJ}$ $\text{C}_2\text{H}_6(\text{g}) + 3\frac{1}{2}\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l}) \quad \Delta H_r^\circ = -1550 \text{ kJ}$ $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l}) \quad \Delta H_r^\circ = -286 \text{ kJ}$	
2.	Illustrate the Hess's cycle for the reaction, $2\text{NaHCO}_3(\text{s}) \rightarrow \text{Na}_2\text{CO}_3(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}).$ Calculate the enthalpy change of this reaction from the following data: $\Delta H_f^\circ(\text{NaHCO}_3) = -840.9 \text{ kJmol}^{-1}, \Delta H_f^\circ(\text{Na}_2\text{CO}_3) = -1130.7 \text{ kJmol}^{-1},$ $\Delta H_f^\circ(\text{CO}_2) = -393.5 \text{ kJmol}^{-1}, \Delta H_f^\circ(\text{H}_2\text{O}) = -286 \text{ kJmol}^{-1}.$	
3.	0.53 g of $\text{C}_4\text{H}_4\text{S}(\text{l})$ is burnt to raise the temperature of a copper made beaker containing 200 g water. The temperature rises from 18.5 to 35.5 °C. Mass of the beaker is 100 g and specific heat for copper and water are 0.385 and 4.18 $\text{Jg}^{-1}\text{C}^{-1}$, respectively. Calculate enthalpy change of combustion.	
4.	A quantity of 200 mL 1.2 molL^{-1} HCl is mixed with 200 mL 0.50 molL^{-1} $\text{Ba}(\text{OH})_2$ in a calorimeter that has a heat capacity of 453 JK^{-1}. The initial temperature of both the solutions before mixing was 20 °C. Calculate the final temperature of the mixed solution. Given that $\Delta H_n = -56.2 \text{ kJmol}^{-1}$ and specific heat for aqueous solution is 4.18 $\text{JK}^{-1}\text{g}^{-1}$.	

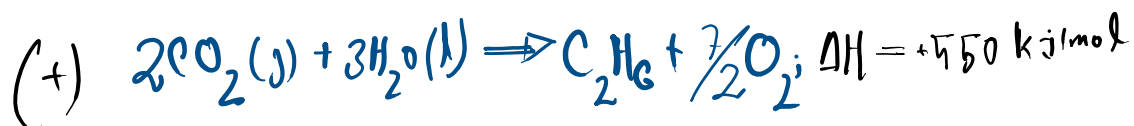
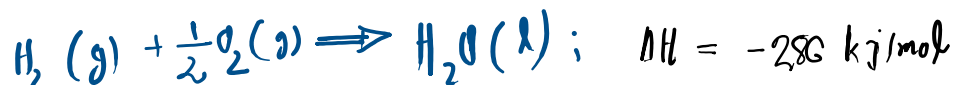
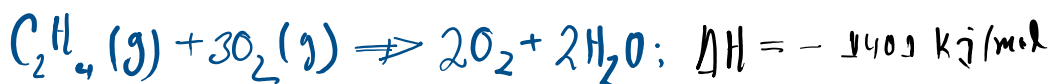
① Solⁿ!

① Soln:

$$\Delta H_r = (-1401 - 286) - (-1150)$$

$$= -137$$

or,



$$\therefore (-1401 - 286 + 1550) = -137 \text{ kJ/mol}$$

② Soln:

$$\Delta H_r = (-1230.7 - 393.5 - 286) - (-2 \times 890.9)$$

$$= -128.4$$

③ Solⁿ:



$$T_2 = 37.5^\circ\text{C}$$

$$T_1 = 18.5^\circ\text{C}$$

$$\Delta T = 19^\circ\text{C}$$

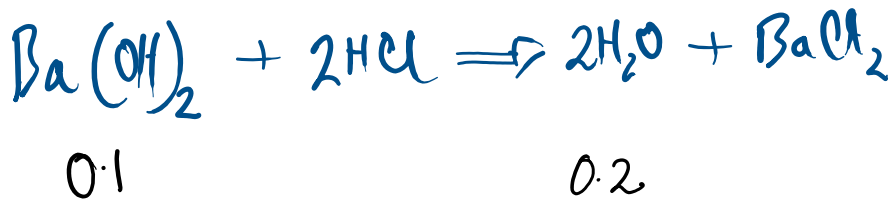
$$\begin{aligned} & n_w c_w \Delta T + n_c c_c \Delta T \\ &= (14212 + 657.5) \text{ kJ mol}^{-1} \\ &= 14869.5 \text{ kJ mol}^{-1} \end{aligned}$$

$$\frac{0.53}{84} \text{ mol produces } 14869.5 \text{ kJ mol}^{-1}$$

$$\therefore 1 \text{ mol } \quad \quad \quad 2.35 \times 10^6 \text{ kJ mol}^{-1}$$

$$= 2356 \text{ kJ mol}^{-1}$$

④ Solⁿ:



$$\begin{aligned} n_{HCl} &= \frac{200 \times 1.2}{1000} \\ &= 0.24 \end{aligned}$$

$$N_{Ba(OH)_2} = \frac{200 \times 0.5}{1000}$$
$$\Rightarrow 0.10$$

$\text{Ba}(\text{OH})_2$ is limiting reagent.

$$\Delta H_v = \frac{Q}{n} \rightarrow \text{of } H_2O$$

$$\text{от, } \Delta H_n = \frac{Q}{0.2}$$

$$\Rightarrow Q = 0.2 \times (-50.2)$$

$$\Rightarrow m, C, \Delta T + (\text{heat capacity} \times \Delta T) = 0.2 \times (+50.2)$$

$$\Rightarrow (400 \times 4.18 \times 117) + (453 \times 117) = (0.2) (+50.2) \times 10^3$$

$$\rightarrow (400 \times 4.18 \times \Delta T) + (453 \times \Delta T) = (0.2) (+50.2) \times 10^3$$

$$\Rightarrow (400 \times 4.18 \times \Delta T) + (453 \times \Delta T) = (0.2) (+50.2) \times 10^3$$

$$\Rightarrow \Delta T = \frac{11200}{(400 \times 4.18) + 453} = 5.27$$

$$\therefore T_2 - T_1 = 5.27^\circ$$

$$\Rightarrow T_2 = 25.27^\circ\text{C}$$